CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a Pythagorean triplet? 1

(a) 10, 13, 15 **(b)** 9, 12, 16 **(c)** 9, 13, 15 **(d)** 9, 12, 15

2. The perimeter of a right triangle with legs 9 cm and 12 cm is (in cm): 1

(a) 37 **(b)** 21 **(c)** 30 **(d)** 36

3. A rectangle is 9 cm long and 12 cm broad. Its diagonal (in cm) is: 1

(a) 15 **(b)** 16 **(c)** 21 **(d)** 14

4. The smallest Pythagorean triplet is: 1

(a) 4, 5, 6 **(b)** 2, 3, 4 **(c)** 3, 4, 5 **(d)** 1, 2, 3

5. The legs of a right triangle are 8 cm and 15 cm. Its hypotenuse (in cm) is: 1

(a) 16 **(b)** 17 **(c)** 18 **(d)** 23

6. The hypotenuse of a right triangle is 17 cm and one leg is 8 cm. The other leg (in cm) is: 1

(a) 9 **(b)** 15 **(c)** 16 **(d)** 14

7. Which of the following is a Pythagorean triplet? 1

(a) 8, 15, 17 **(b)** 8, 15, 18 **(c)** 9, 16, 17 **(d)** 8, 16, 17

8. The perimeter of a right triangle with legs 8 cm and 15 cm is (in cm): 1

(a) 41 **(b)** 40 **(c)** 34 **(d)** 23

9. A rectangle is 8 cm long and 15 cm broad. Its diagonal (in cm) is: 1

(a) 23 **(b)** 16 **(c)** 17 **(d)** 18

10. The Baudhayana–Pythagoras theorem is true for: 1

(a) equilateral triangles only **(b)** isosceles triangles only
(c) all triangles **(d)** right-angled triangles only

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Is a triangle with sides 9, 12, 15 cm right-angled? 2

12. The legs of a right triangle are 9 cm and 12 cm. Find its hypotenuse. 2

13. In a right triangle the hypotenuse is 25 cm and one leg is 15 cm. Find the other leg. 2

14. Is a triangle with sides 5, 7, 9 cm right-angled? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 17 m long leans against a wall. Its foot is 8 m from the wall. How high on the wall does it reach? 3

16. Find the perimeter of a right triangle whose legs are 12 cm and 16 cm. 3

17. Rohan walks 7 km east and then 24 km north. How far is he from his starting point? 3

OR

A ladder 25 m long leans against a wall. Its foot is 7 m from the wall. How high on the wall does it reach? 3

18. Find the perimeter of a right triangle whose legs are 15 cm and 20 cm. 3

OR

Rohan walks 12 km east and then 16 km north. How far is he from his starting point? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular field is 8 m long and 15 m broad. Find the length of its diagonal. A man walks along the diagonal instead of the two sides; how much distance does he save? 5

OR

Using $(2m,\ m^2-1,\ m^2+1)$ with $m = 6$, write a Pythagorean triplet and verify it. 5

20. A rectangular field is 7 m long and 24 m broad. Find the length of its diagonal. A man walks along the diagonal instead of the two sides; how much distance does he save? 5

OR

Using $(2m,\ m^2-1,\ m^2+1)$ with $m = 7$, write a Pythagorean triplet and verify it. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND